MACHINE LEARNING ASSIGNMENT – PREDICT INSURANCE CHARGES

1.PROBLEM IDENTIFICATION:

Machine Learning – Because need to predict numbers

Supervised learning – Because Requirement, Input and Output are Available

Regression – Because need to predict Numerical Value.

2.DATASET:

- Name - insurance_pre.csv
- 1338 rows × 6 columns

3.ALGORITHMS CHECKED:

Starts from Multiple Linear Regression – Because there are Multiple Inputs here and single output required – Simple Linear Regression not applicable.

- Multiple Linear Regression
- Support Vector Machine
- Decision Tree
- Random Forest

3.RESULTS:

1.Multiple Linear regression r value = 0.78947903498670009.

2.Support Vector Machine

S.NO	Hyper parameter	Linear (r value)	RBF (Non Linear)(r value)	Poly(r value)	Sigmoid(r value)
1	C=10	-0.0016176	-0.081969104	-0.093116155	-0.0907832
2	C=100	0.54328182	-0.124803678	-0.099761723	-0.11814555
3	C=500	0.62704628	-0.124641613	-0.082028799	-0.45629443
4	C=1000	0.63403693	-0.117490924	-0.055505938	-1.66590813
5	C=1500	0.63942118	-0.112389909	-0.028732414	-3.31637406
6	C=2000	0.68932631	-0.10778764	-0.002702451	-5.61643154
7	C=2500	0.7135352	-0.102426338	0.023441363	-8.48996266
8	C=3000	0.75908904	-0.096212851	0.048928964	-12.0190481

Best r value - SVM Linear = 0.75908904

3.Decision Tree Regression

S.NO	CRITERION	MAX FEATURES	SPLITTER	R VALUE
1	friedman_mse		best	0.688445004
2	absolute_error		best	0.675956068
3	squared_error		best	0.705866285
4	friedman_mse		random	0.757129588
5	absolute_error		random	0.71865612
6	squared_error		random	0.730376173
7	friedman_mse	sqrt	best	0.698080274
8	absolute_error	sqrt	best	0.69437879
9	squared_error	sqrt	best	0.725790424
10	friedman_mse	log2	best	0.701411309
11	absolute_error	log2	best	0.714270525
12	squared_error	log2	best	0.688416092
13	friedman_mse	sqrt	random	0.653732324
14	absolute_error	sqrt	random	0.758188088
15	squared_error	sqrt	random	0.694624315
16	friedman_mse	log2	random	0.6168899
17	absolute_error	log2	random	0.686573672
18	squared_error	log2	random	0.666195338

Best r value – Decision Tree Regression – absolute_error , sqrt , random = 0.758188088

4.Random Forest

S.NO	CRITERION	MAX FEATURES	SPLITTER	R VALUE
1	Default		n estimator=100	0.853552161
2	friedman_mse	sqrt		0.869222063
3	absolute_error	sqrt		0.872730005
4	squared_error	sqrt		0.871635983
2	friedman_mse	log2		0.870779564
3	absolute_error	log2		0.873907133
4	squared_error	log2		0.870457151

Best r value – Random Forest – absolute_error , log2 = 0.873907133

4.Final model Selection

Highest validation of the r value - Random Forest – absolute_error , log2 = 0.873907133 compared to alternatives.